

Research article

Farmers perceive diminishing ecosystem services, but overlook dis-services in intensively used agricultural landscapes in the North China Plain

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ABSTRACT

The UN sustainable development goals ask countries to advance sustainable production methods in agriculture. While the need for a transition to sustainable agricultural production is widely felt, there is little insight into local stakeholders' perceptions regarding agroecosystem (dis)services in areas with intensive production methods. The North China Plain is an agricultural production area with intensive production systems and simplified agricultural landscapes. We conducted a survey with 267 farmers in Quzhou county in the North China Plain in 2020 to measure the perceived level of agroecosystem (dis)services supply and the changes therein between 2015 and 2020. We analyzed which explanatory factors were associated with farmers' perceptions. Provisioning services were at a high level, while the regulating and supporting ecosystem services were considered to be in low supply, as evidenced by low scores for the presence of natural enemies and earthworms, and for natural habitats such as hedgerows and windbreaks. Most of the participants did not perceive dis-services from agriculture. Differences in perception between villages with contrasting biophysical and socio-economic conditions highlight the relevance of contextualized policy development for agricultural landscape composition and configuration to manage ecosystem (dis)services.

1. Introduction

The global call for action on United Nations' 2030 Sustainable Development Goals (SDGs) emphasizes the urgent need for environmental sustainability and responsible resource management, especially in agricultural systems (DeClerck et al., 2016; UN General Assembly, 2015). The global concerns have triggered various initiatives in China. In 2016, the China central government's No. 1 Document proposed the concept of "Agriculture Green Development" as a point of departure for sustainable intensification in agricultural landscapes (State Council, 2016). This was made actionable with the Rural Revitalization Strategy

of 2017 and the "14th Five-Year National Agriculture Green Development Plan" published in 2021 (Ministry of Agriculture and Rural Affairs, 2021). One of the main goals of *Agriculture Green Development* is to deliver balanced multiple agroecosystem services, reduce dis-services and improve human well-being (Shen et al., 2020). Assessment of the supply of multiple agroecosystem (dis)services is the benchmark towards integrating ecosystem services into agricultural landscape planning, management, and decision-making (de Groot et al., 2010).

We zoomed in on the North China Plain as a case study area because it is an intensive crop production area that is critical to China's food security, producing approximately a quarter of China's grain (NBSC,

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2019). Double cropping of winter wheat (grown from October to June) and summer maize (from June to October) is one of the main cropping systems in the North China Plain. This cropping system has high yearly grain output, but has relatively low performance regarding regulating, supporting, and cultural services (Zhou et al., 2017). Moreover, there is a high risk of dis-services, for instance, water pollution by nitrogen losses (Meng et al., 2022), high greenhouse gas emissions (Zhao et al., 2021) and groundwater table decline (Zhang et al., 2003).

Previous work, using the methods of emergy (Odum, 1996) and life cycle assessment has indicated unsustainable agroecosystem conditions in the North China Plain (Cui et al., 2021; Ma et al., 2015). However, key scientific information is missing on essential multiple agroecosystem (dis)services at the landscape level like biological control and cultural services. Conducting quantitative biophysical assessment on multiple ecosystem (dis)services is costly and time-consuming, and survey methods offer an attractive low-cost alternative. For cultural services, surveys might even be a more appropriate instrument to collect the information because such services are intrinsically based on stakeholder perceptions. It stands to reason to ask local actors about the current state and recent historic trends in agroecosystem services and dis-services in their region, to capitalize on local knowledge, which can be accomplished with relatively limited costs for a broad suite of services.

Another advantage of using stakeholder surveys is that an agricultural green development transition can only be realized with the support of local stakeholders. Hence, it is necessary to also explore farmers' perceptions about the main agroecosystem (dis)services and the temporal change in their supply. Farmers shape agroecosystem (dis)services as decision-makers, influenced by landscape features and policies (de

Snou et al., 2013). Their perceptions, reflecting local knowledge, complement biophysical measurements, particularly for cultural (dis)services (Dawoe et al., 2012; de Groot et al., 2002). Perception-based surveys have the potential to efficiently gather local-specific information on agroecosystem (dis)services (Chen et al., 2017; Liu et al., 2022), aiding policymakers and managers in implementing context-specific policies (Vrebos et al., 2015).

Villages in China vary in ecological and social-economic conditions (Wang et al., 2021), which may impact agroecosystem (dis)services (Yang and Wang, 2022) and eventually rural development (Cao et al., 2020; Domon, 2011). Socioecological conditions and landscape patterns affect agroecosystem (dis)services and thus people's perceptions (Wang et al., 2021; W. Zhang et al., 2016; Zoderer et al., 2019). While agricultural policies are typically implemented at the village level, there is little information on contextual factors at the village level for enhancing agroecosystem (dis)services in the North China Plain.

In intensively managed agricultural landscapes, we expect that the regulating and supporting services would be low and there should be signs of dis-services. We also expect that farmers' perceptions could be associated with socio-economic factors and biophysical characteristics of villages. In this paper, we aim to assess the perception of farmers regarding the delivery of agroecosystem services and dis-services in a landscape dominated by intensive cropping systems in Quzhou county within the North China Plain. We quantified the perceived supply of these services and the changes therein during 2015–2020. To investigate how village characteristics affect the perception of agroecosystem (dis)services, we further analyzed the relations between explanatory factors and multiple agroecosystem (dis)services. Finally, we compared

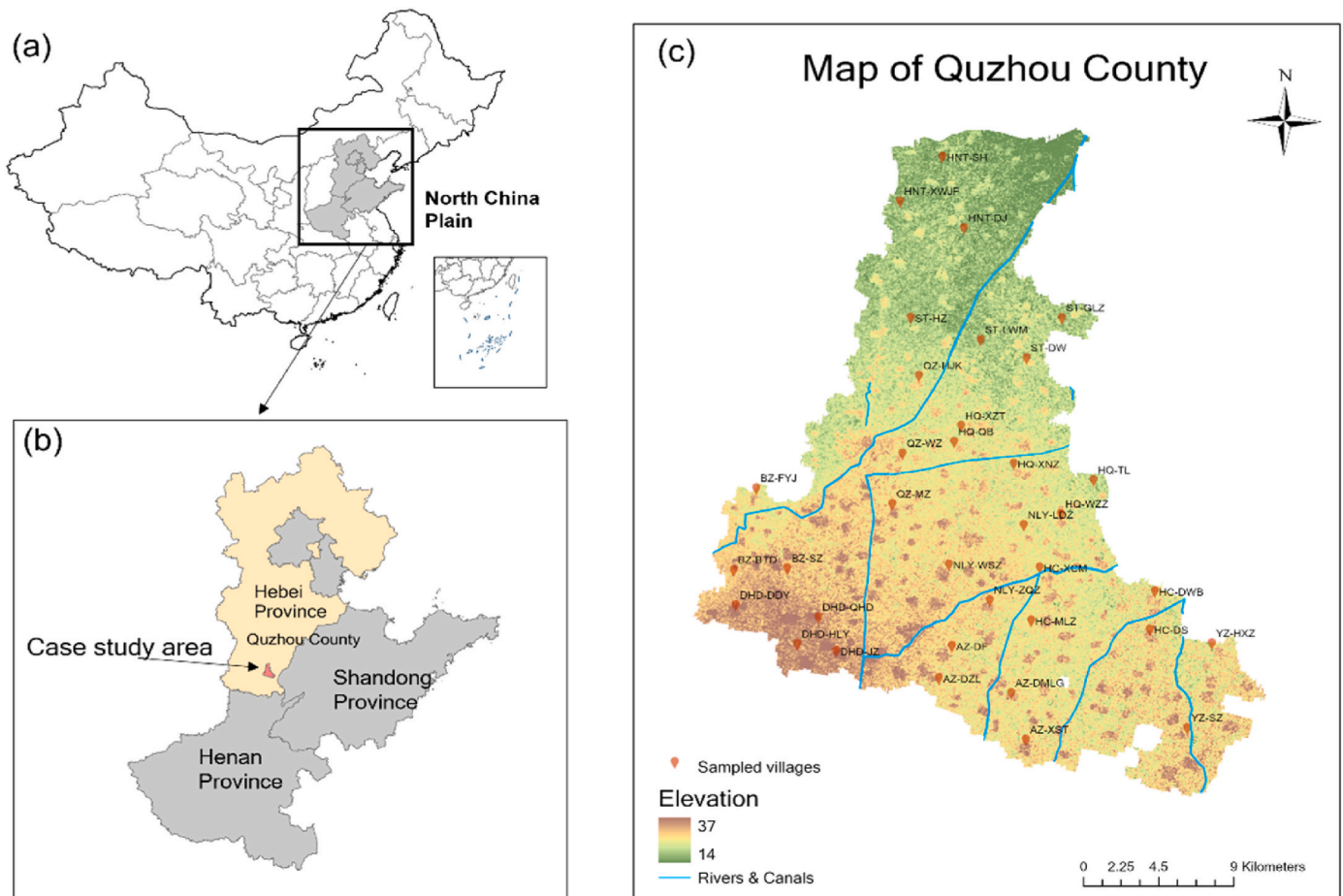


Fig. 1. Location of the North China Plain (a) and Quzhou county (b) and of sampled villages within Quzhou County (c). The elevation is derived from 12.5 m Digital Elevation Model (DEM) topographic data (<https://search.asf.alaska.edu/>).

agroecosystem (dis)services in different village types.

2. Materials and methods

2.1. Case study area

The case study area is Quzhou county, an affiliated administrative region of Handan City, Heibei Province, located in the middle of the North China Plain (Fig. 1). As a representative area in the North China Plain, Quzhou has been selected as a National Agriculture Green Development Pilot Demonstration Zone in 2019. Since 2020, the Quzhou government has formulated policies to transform the county towards sustainable agricultural development in the context of the Agricultural Green Development program.

The surface area of Quzhou is 676.7 km² including 525.2 km² of cropland (NBSC, 2017). The landscapes in Quzhou County are for the most part homogenized agricultural landscapes with arable cropping and little area of non-crop habitat. The soil in Quzhou is calcareous alluvial with loamy texture, high soil pH, relatively low soil organic matter, and historically high salt content (K. Li et al., 2022; Liang et al., 2022). There are three main rivers (Fuyang River, Zhizhang River, and Laosha River). Quzhou is located in a semi-humid continental monsoon climate zone, with an average annual temperature of 13.1 °C and an average annual average precipitation of 534.9 mm, while most precipitation occurs between July and September. Winter wheat, summer maize and cotton are the main crops. The tree cover area in Quzhou county is 86.7 km², mainly consisting of linear windbreaks (Li, 2014; NBSC, 2017). Since 2015, Quzhou has enacted a fallow policy that discourages the winter wheat-summer maize cropping system to save groundwater. Around 2300 ha of winter wheat-summer maize system was replaced by spring maize, spring cotton or other water-saving systems (Quzhou County Government, 2020).

Quzhou County consists of 10 townships and 342 administrative villages. According to NBSC (2017), 85.5% of the population in Quzhou County is rural. The total agricultural revenue amounted to 1.03 billion CNY 2017 (around 0.15 billion USD), which comprises 7.6% of the county's total economic output.

2.2. Methodological framework

The ecosystem services and dis-services framework used in this study

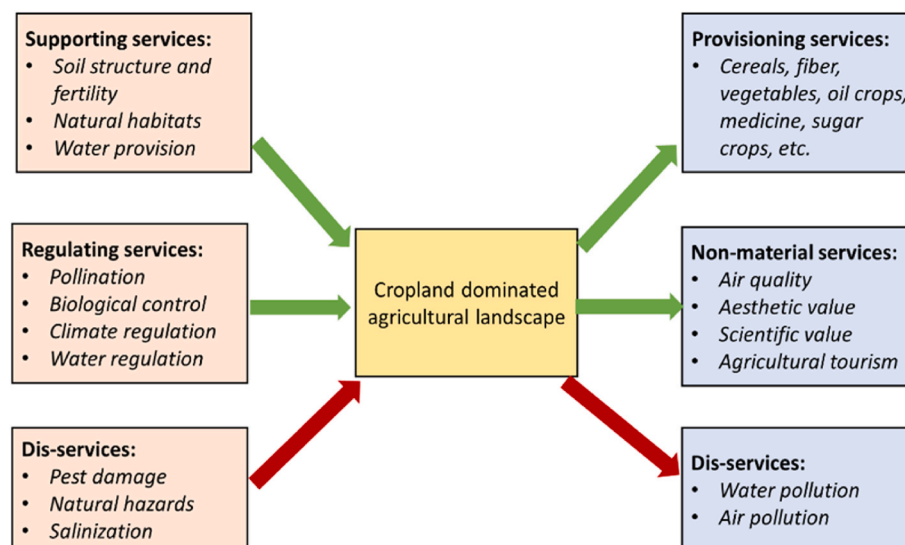


Fig. 2. Main interrelations between agroecosystem services and dis-services, and crop production in the agricultural landscape of Quzhou county. Agroecosystem (dis)services in orange boxes affect the agricultural landscape, while services and dis-services in blue boxes are affected by agriculture. The green arrows indicate services, whereas the red arrows indicate dis-services (adapted from Zhang et al., 2007).

(Fig. 2) is based on the four common categories of provisioning services, regulating services, supporting services, and cultural services (MEA, 2005). The framework is adapted to include dis-services (Zhang et al., 2007) as well as locally relevant interactions between agroecosystem (dis)services and the agricultural landscape (Fig. 2). Crop production benefits from supporting and regulating services, but can be negatively affected by dis-services. In turn, the cropping activities result in provisioning services and contribute to non-material services, but they also deliver dis-services to the agroecosystem. This framework and a literature review were used to create a list of proxy indicators for relevant agroecosystem (dis)services and to formulate a questionnaire to assess perceptions of the state of agroecosystem (dis)services in 2020 and the perceived trend of change in the period between 2015 and 2020.

The survey targeted farmers within 35 villages in 2020. The selection of villages was based upon a pre-defined village typology in order to cover the diversity of villages (see Section 2.3). A redundancy analysis (RDA) was done to investigate relationships between village characteristics and the perceived supply of agroecosystem (dis)services in 2020. Finally, we analyzed differences in agroecosystem (dis)services among village types.

2.3. Selection of villages and farmers

Participants were selected using a stratified sampling approach, based on a village typology (see Supplementary Materials S2). This typology classified the villages of Quzhou county into eight different types based on biophysical, socioeconomic, and agricultural production characteristics from previous survey data and yearbook data. The seven village types which had agricultural production were selected for further research (thus excluding the village type without agricultural production), and five villages were randomly selected from each village type, resulting in 35 selected villages in total. In each village, we selected for participation in the survey five smallholder farmers, up to two leader farmers and the village head. The smallholder farmers had experience in crop production and were randomly selected per village. Leader farmers are defined as farmers running comparatively professional farms, or had participated in professional farmers' training. If no leader farmers who met the criteria were available, we interviewed additional smallholder farmers instead (coded as smallholder farmers). Leader farmers were identified with the help of the village heads. The 35 village heads were also participating in agricultural production, so that all survey

Box 1

The terms used in this paper.

- **Agroecosystem services and dis-services:** Ecosystem services are “the benefits people obtain from ecosystems” (MEA, 2005). They entail the products (for profit) or the (non)material goods (for use) that people gain from agroecosystems. Dis-services are negative impacts on agroecosystems or humans.
- **Agroecosystems:** Ecosystems used for agricultural production.
- **Ecosystem services supply:** The provisioning of a specific bundle of ecosystem goods and services within a given time period in an area.
- **Perceived ecosystem services supply:** Flow of ecosystem (dis)services from certain ecosystems (provider) as experienced by people (receiver).
- **Trend of ecosystem service supply:** Temporal changes of ecosystem services supply during the time period of 2015–2020.

respondents can be regarded as farmers. In total, 267 people completed the survey, including 187 smallholders, 45 leader farmers and 35 village heads.

2.4. Survey

The survey was conducted in December 2020 using a structured questionnaire, consisting of three parts: 1) demographic and personal information of participants, e.g., age, gender, and education; 2) crop production information such as total planting area of each crop, annual average revenue (price times production) of each crop; and 3) perceptions of agroecosystem services and dis-services. Perceptions were quantified by scoring proxy indicators of supply and changes in the period between 2015 and 2020 on a Likert Scale from 1 to 5 (Table S1). Due to the length of the questionnaire, we were not able to ask for all kinds of costs explicitly for calculating the net revenue. Therefore, we had to rely on gross revenue, ignoring the costs of planting crops, which is an obvious limitation of our study. The questionnaire was tested through 3 rounds of pilot surveys with local experts and farmers from October to November 2020. This resulted in elimination of indicators that were considered not locally relevant, such as soil erosion. During the pilot surveys, we iteratively revised the questionnaire to ensure that the questions would be understandable for the respondents. During the interview, the meaning of each ecosystem (dis)service was explained. The interviewer team of 9 persons was trained before going into the field and had background knowledge of survey methods and familiarity with the local dialect. After the entire survey was completed, the quality of

each questionnaire was examined. Details of the questionnaire indicators used in this study are listed in Table S1 in Supplementary Materials.

2.5. Data on explanatory factors

Multiple data sources were used to describe the characteristics of each village (Table 1).

2.6. Data analysis

Indicator scores were visualized using the “Likert” (Bryer et al., 2016) and “ggplot2” packages (Wickham et al., 2016) in R (R Core Team, 2021). To analyze how village-level characteristics were associated with the perceived agroecosystem services and dis-services supply in 35 villages, a redundancy analysis (RDA) was conducted using village-averaged indicator scores and village-specific biophysical and socio-economic variables. RDA is a widely used analysis to explore the correlation between multiple response variables (agroecosystem services and dis-services supply in this case) and a set of explanatory variables (socio-ecological factors in this case). Prior to the RDA, the indicator values of agroecosystem (dis)services supply and variables representing village characteristics were standardized to zero mean and unit variance. A Monte Carlo permutation test (1000 permutations) was performed to determine the significance of the associations between village characteristics and (dis)services. The RDA and the data transformation were done using the “vegan” package in R (Oksanen et al.,

Table 1

Explanatory factors describing the characteristics of villages involved in the survey.

Factors	Unit	Data sources and/or calculation method
Soil properties		Soil survey data in 2018 (Li et al., 2022b)
SOM (soil organic matter)	g/kg	Inverse distance weighted method to calculate the village average in ArcMap 10.8. (https://www.esri.com/en-us/arcgis/products/arcgis-desktop)
TN (soil total nitrogen)	g/kg	
EC (electrical conductivity)	dS/m	
Agricultural inputs		Household survey data in 2020 (Liang et al., 2022).
Fertilizer use amount	kg N ha ⁻¹ year ⁻¹	See Section S5 in Supplementary Materials for the calculation method.
Pesticide use frequency	TFI	Calculation method is based on Liang et al. (2022)
Main crop type proportion		Village head survey data, 2020
Cereals	%	Winter wheat-Summer maize area or spring maize area/total crop production area
Fiber crops	%	Cotton area/total crop production area
Fruit trees	%	Fruit trees area/total crop production area
Sugar crops	%	Stevia area/total crop production area
Vegetables	%	Vegetables area/total crop production area
Location advantages		LULC map and administrative map
Distance to the county center (Quzhou city)	km	Methods of spatial join (closest) in ArcGIS Pro 2.8.0 (https://support.esri.com/en/products/desktop/arcgis-desktop/arcgis-pro/2-8)
Distance to the nearest river	km	
Socio-economic factors		Village head survey data in 2018. See Section S4 for more information in Supplementary Materials.
Non-agricultural labor	%	Yearbook data (NBSC, 2017)
Population	Capita	

2013).

We used the parameter-free Kruskal-Wallis test to compare agroecosystem (dis)services supply among the different village types. If the overall Kruskal-Wallis test was significant, comparisons between two village types were performed with Wilcoxon Rank Sum Tests. These parameter free tests do not assume any particular distribution for the data and do not require a normality assumption. The mean scores of the agroecosystem services and dis-services were plotted as bubble plots using the package 'ggplot2' (Wickham et al., 2016) in R (R Core Team, 2021).

A recoding of indicators was done for the RDA and Kruskal-Wallis analysis. In the case of natural enemies (frogs, ground beetles, ladybeetles), water sufficiency (river water, deep well water and shallow well water), semi-natural habitats, and dis-services to agriculture, we treated the "No occurrence" or "Not applicable" responses as a true zero observation (no presence of this service or disservice). Furthermore, for disservices to or from agriculture, if respondents provided a score, we recoded the Likert scores 1–5 into a single rating of 1 (dis-service present). See Table S1 in Supplementary Materials for detailed scales after such recoding. If the farmers do not have any insect-pollinated crops, they were excluded from the analysis. If respondents gave a score of "Don't know" (seen as the true missing value) in the cases of cultural services, soil moisture, air quality and temperature change (in total 2% of the respondents), we used for the RDA and Kruskal-Wallis analysis the village average as a response for these respondents, assuring that in these cases the response would not affect the calculated mean (Roth, 1994).

3. Results

3.1. Provisioning ecosystem services

The agricultural landscape of Quzhou is dominated by five crop types: cereals (mostly winter wheat, *Triticum aestivum*, spring and summer maize, *Zea mays*), vegetables (cabbage, *Brassica oleracea* var. *Capitata*, chillis, *Capsicum frutescens*, etc.), fruit trees (apple, *Malus domestica*, pear, *Pyrus L.*, etc.), sugar crops (*Stevia*, *Stevia rebaudiana*), and fiber crops (cotton, *Gossypium hirsutum*). Vegetables and fruit trees had the highest average annual revenue, of 73,000 and 87,700 CNY per ha, respectively, although the variation was large (Table 2). More than half of the respondents perceived a reduction (scored as 'weak decline' or 'strong decline' in Table 2) in the revenue from fiber and sugar crops over the last 5 years, while fruit trees and vegetables were considered by most farmers to have become more remunerative.

3.2. Regulating ecosystem services

Most participants were aware of the state of regulating services and the changes therein (Fig. 3 c and d). Between 51% and 79% of the participants scored the presence of natural enemies (ladybugs, frogs, ground beetles) as 'low' to 'very low' (Fig. 3a), and 61–75% of the respondents perceived a 'weak decline' or 'strong decline' in the numbers of these organisms in the period 2015 to 2020 (Fig. 3b). Most farmers (64%) scored the soil moisture condition as 'poor' to 'very poor', but

47% observed an improvement from 2015 to 2020 (Fig. 3). Half of the farmers were planting insect-pollinated crops (Fig. 3c), and forty percent of them thought the effectiveness of pollination was 'poor' to 'very poor' (Fig. 3a). Forty-seven percent of the respondents perceived an increase in temperature from 2015 to 2020 (Fig. 3a). More than 80% of the people assessed the air quality as good and improving (Fig. 3). In sum, some services were provided at an adequate level (air quality) while others were low (frogs, ground beetles, ladybugs, and soil moisture) while some were intermediate (pollination). Frogs, ladybugs, and ground beetles showed a declining trend, while air quality and soil moisture were improving, and pollination was stable. As a robustness check, we analyzed our data also with only smallholder farmers (excluding village heads and leader farmers) and obtained very similar patterns (see Fig. S5). However, village heads had on average higher ratings than the other interviewees for pollination (Table S5).

3.3. Supporting ecosystem services

The scorings provided by the participants indicate a shortage of supporting ecosystem services (Fig. 4). A minority of the participants indicated that natural habitats (windbreaks, wildflowers, hedgerows, field islets, diversified cropping, and ditches) were present in their villages (Fig. 4c). Where these elements were present, less than half of the respondents thought the presence of natural habitats was abundant as indicated with a score of 'high' or 'very high' (Fig. 4a). The abundance of earthworms, as an indicator of soil fertility and biodiversity, was perceived as 'low' or 'very low' by 64% of participants (Fig. 4a), and more than half of the participants thought the number of earthworms showed a 'weak decline' or 'strong decline' between 2015 and 2020 (Fig. 4b). More than half of the participants perceived a declining groundwater sufficiency (both in shallow wells and deep wells) between 2015 and 2020 providing a score of 'decline' or 'strong decline'. On the other hand, fifty-nine percent of the participants perceived an improvement in the river water supply, by scoring either 'weak increase' or 'strong increase' (Fig. 4b). Thus, the state of half of the supporting services (length of hedgerows, earthworms, windbreaks, diversified cropping systems, wild flowers) was rated by the majority of respondents as low or very low, but only three services were rated by a majority as declining (earthworms, and deep and shallow wells) while for other services no strong tendency to decline was indicated. The results are mostly similar across different farmer groups, though smallholders tend to report less sufficient river waters, fewer diversified cropping systems, and less rich field islets than leader farmers and village heads while village head gave higher scores of those services (see Table S5).

3.4. Cultural ecosystem services

Most participants (58%) perceived the beauty of the landscape as a cultural service and 44% scored the suitability for agrotourism as 'good' to 'very good' (Fig. 5a). Improvements in these cultural services occurred in the years 2015–2020, as 63% and 66% of the respondents scored 'increased' or 'strongly increased' for these indicators (Fig. 5b). The involvement in science and education was rated as 'low' to 'very

Table 2
Provisioning services supply expressed in revenue of different crop types in 2020 and perceived trend in revenue between 2015 and 2020.

Crop Types	Crop revenue (Unit: 10 ³ CNY/ha, 1 CNY ≈ 0.15 USD)			Trend of crop revenue from 2015 to 2020 (Percentage of the respondents)				
	N ^a	Mean (±sd)	Range	Strong decline	Weak decline	Stable	Weak increase	Strong increase
Cereals	459	17.7 (±5.6)	5.3–97.0	7.2%	19.2%	30.7%	32.7%	10.2%
Fiber crops	140	23.3 (±6.4)	0.2–47.9	22.9%	40.7%	17.1%	15.7%	3.6%
Vegetables	128	73.0 (±75.4)	3.6–450.0	7.8%	17.2%	19.5%	36.7%	18.8%
Sugar crops	36	37.1 (±14.9)	13.5–66.8	30.6%	22.2%	16.7%	19.4%	11.1%
Fruit trees	19	87.7 (±52.2)	9.1–150.0	10.5%	10.5%	15.8%	47.4%	15.8%

^a N is the number of records for each crop type.

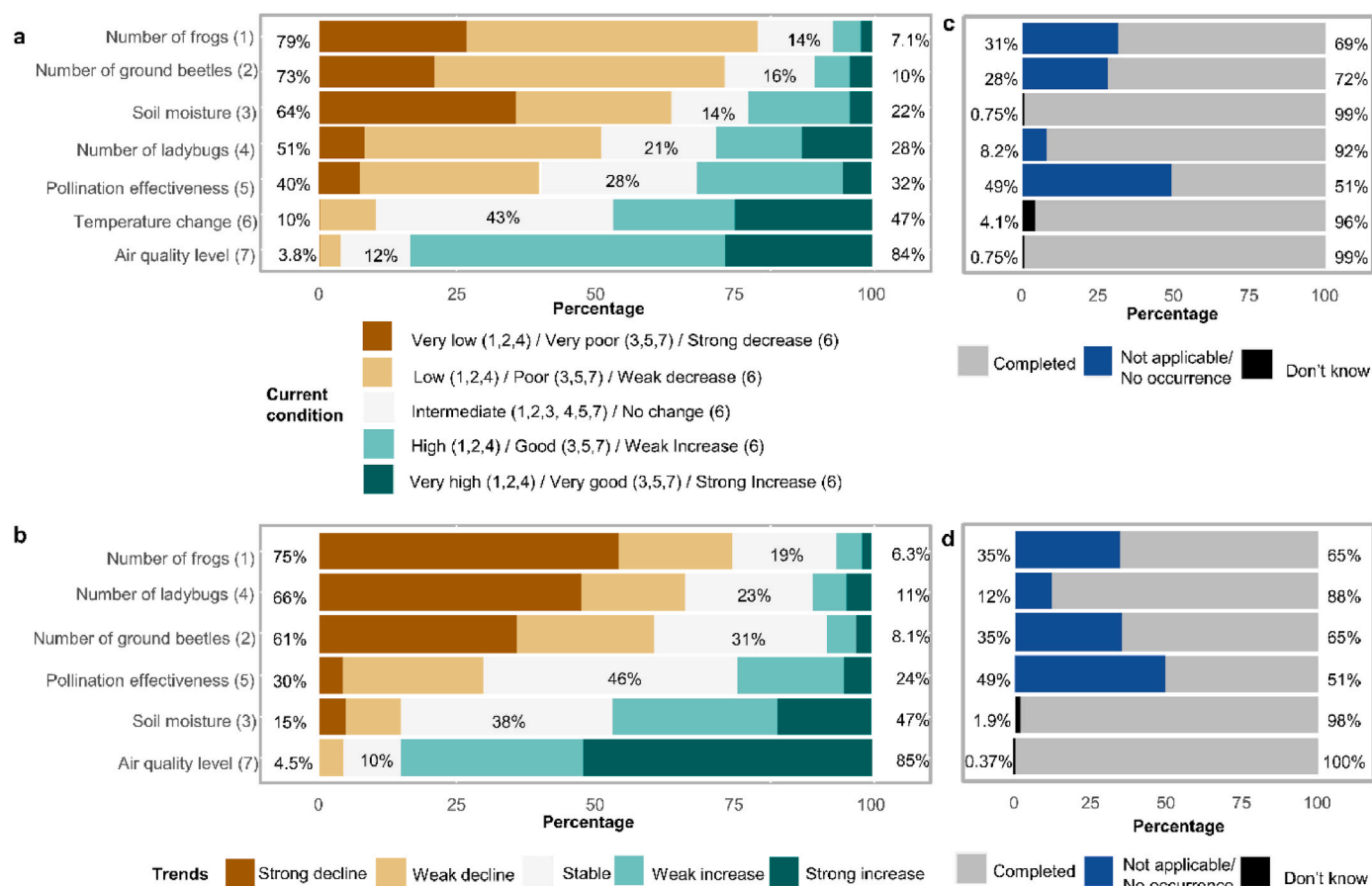


Fig. 3. Perceived supply in 2020 (Fig. 3a) and changes in supply between 2015 and 2020 (Fig. 3b) of regulating ecosystem services. Results in panels a and b are for the respondents who provided a scoring, while panels c and d present the percentage of respondents not providing a scoring. Note: 49% of farmers did not have insect-pollinated crops and hence did not provide a score for pollination effectiveness. See Table S1 in supplementary material for further information about the indicators used in the questionnaire.

low' by 79% of the participants (Fig. 5a), but 33% mentioned improvement in this (Fig. 5b). Awareness about cultural services was relatively high (Fig. 5 c and d). The beauty of the landscape and scientific value was rated significantly lower for smallholders compared to village heads and leader farmers (see Table S5).

3.5. Dis-services

Two types of dis-services were distinguished: dis-services that affect agricultural production (natural hazards and salinization), and dis-services generated by agricultural production (eutrophication, water pollution, and air pollution). The latter were observed by only 14–22% of the farmers (Fig. 6c), and if these were observed, the severity was considered to be mostly 'mild' or 'very mild' (Fig. 6a). Natural hazards (except hail) were reported to have caused yield damage according to more than 40% of the participants (Fig. 6c). Sixty-three percent of the participants claimed that waterlogging was severe or very severe and fifty-six percent rated salinization as severe or very severe (Fig. 6a). Pest damage was not considered to be problematic with 83% of participants scoring 'mild' or 'very mild' effects. Most respondents (ranging from 54% for water logging to 93% for pest damage scoring 'strong decline', 'decline' or 'no change') indicated that impacts of natural hazards to agriculture did not become worse from 2015 to 2020 (Fig. 6b). Three of the most serious natural hazards indicated by respondents were thus related to extreme weather (waterlogging, heat and drought) while salinization was also frequently indicated. Ratings for trend indicate that there are more dis-services with an improving trend than dis-services with a trend towards more serious dis-service. Smallholders were less

likely to report air pollution and salinization than village heads and leader farmers while village heads had higher scores for those dis-services. The village heads also had overall lower scores for several dis-services of natural hazards, such as coldness, drought, dry-hot wind (see Table S5).

3.6. The relations between agroecosystem (dis)services supply and explanatory factors

We used RDA to assess associations between village characteristics and the supply of agroecosystem services and dis-services. The associations between the agroecosystem (dis)services supply and explanatory factors were significant (p -value < 0.001) in permutation tests, and 21% of the total variance in agroecosystem (dis)services supply was explained by the selected explanatory factors (Fig. 7). Among all the variables, the crop area proportion (Vegetables, Fiber crops, Sugar crops, Fruit trees) and distance to the city center (Distance to city) had the most impact on agroecosystem (dis)services supply.

The first axis of the RDA explained 11.7% of the variance (Fig. 7) and was positively correlated with the area proportion of cotton (Fiber crops) but negatively correlated with the area proportion of vegetables. Areas with more vegetables were closer to the city center and rivers (Distance to city, Distance to rivers). The area proportion of vegetables (Vegetables) was positively correlated with higher fertilizer and pesticide application (variables N_input and TFI), higher soil organic matter content (SOM), and electrical conductivity (EC) and it had a positive relation with some of the provisioning services, supporting and regulating services (Revenue, Earthworms, Pollination, and Shallow well

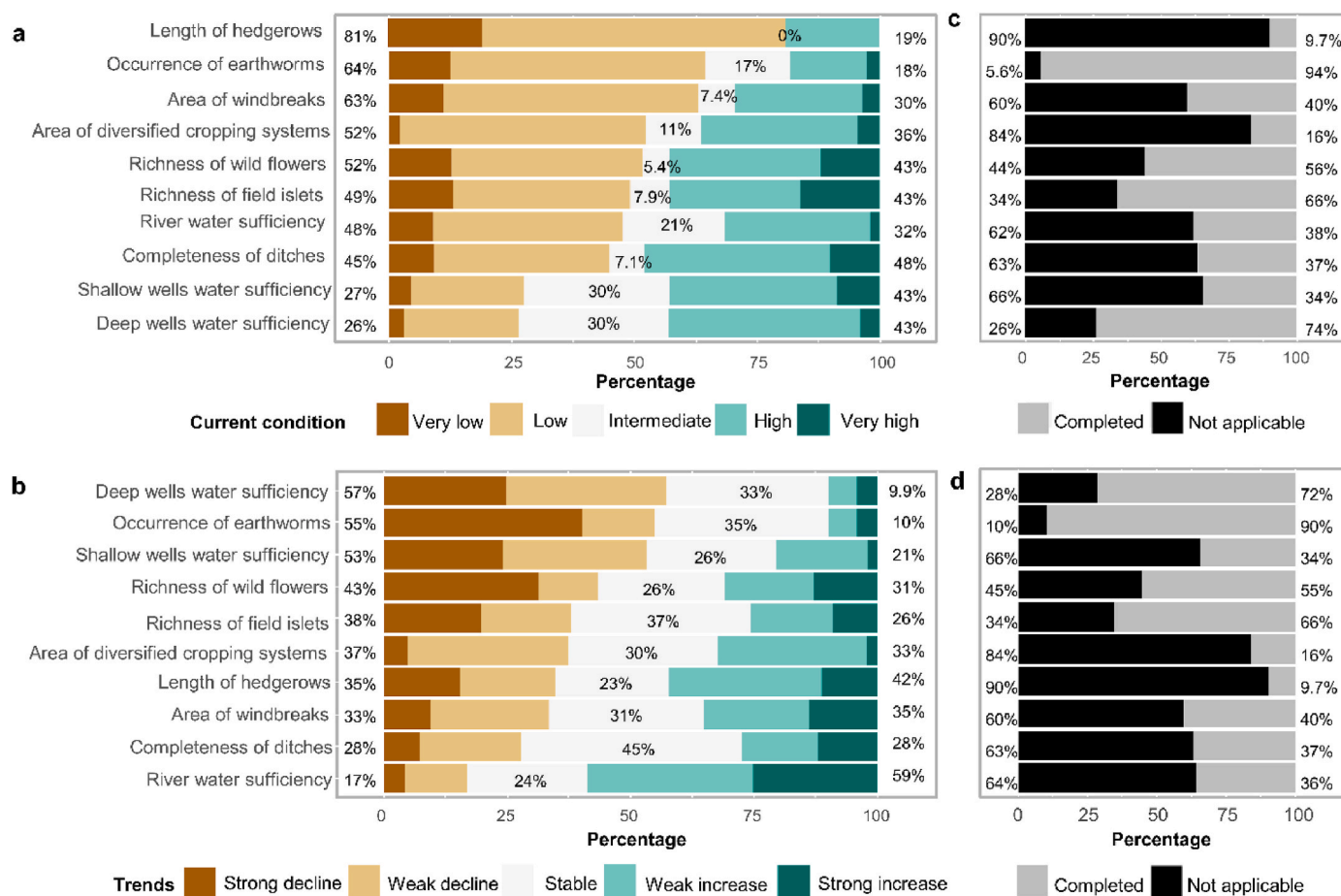


Fig. 4. Perceived supply in 2020 (Fig. 4a) and changes in supply between 2015 and 2020 (Fig. 4b) of supporting ecosystem services. Results in panels a and b are for the respondents who provided a scoring, while panels c and d present the percentage of respondents not providing a scoring. “Not applicable” denotes the absence of specific ecosystem services in village or farm level. For example, some of the villages had no rivers nearby or the farmers did not use certain water resources for irrigation.

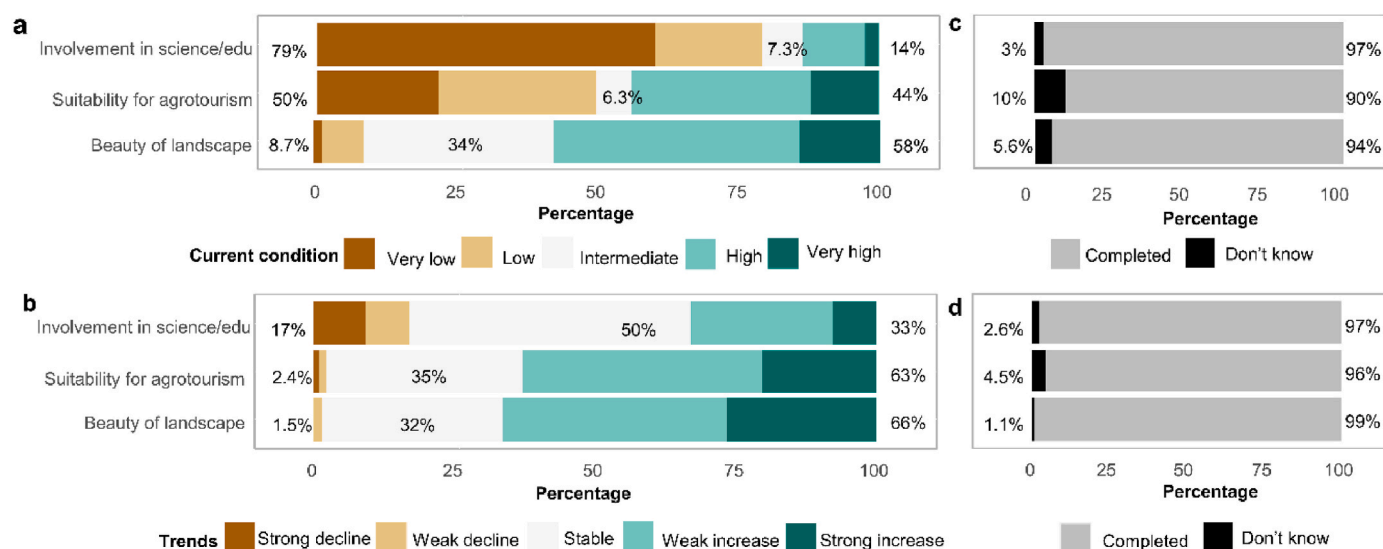


Fig. 5. Perceived supply in 2020 (Fig. 5a) and changes in supply between 2015 and 2020 (Fig. 5b) of cultural ecosystem services. Results in panels a and b are for the respondents who provided a scoring, while panels c and d present the percentage of respondents not providing a scoring.

water) (Fig. 7). These features were negatively correlated with the area proportion of cotton (Fiber crops). In addition, the area percentage of cotton is also positively related to greater severity of dis-services (Pests,

Drought, etc.).

The second axis of the RDA explained 9.4% of the variance (Fig. 7) and was negatively related to the area ratio of Stevia (Sugar crops) and

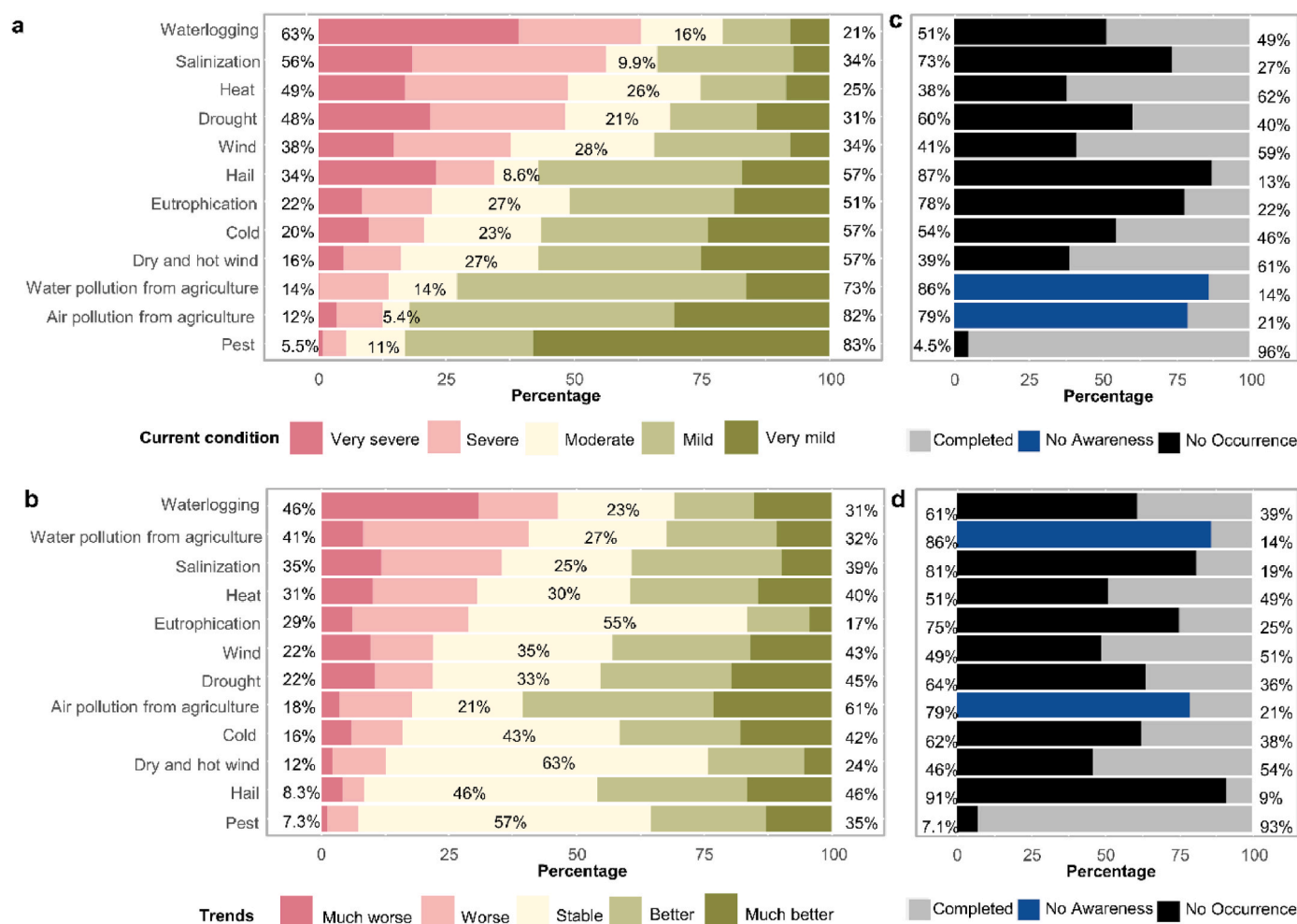


Fig. 6. Perceived damage level in 2020 (Fig. 6a) and changes in the damage between 2015 and 2020 (Fig. 6b) of ecosystem dis-services. Results in panels a and b are for the respondents who provided a scoring, while panels c and d present the percentage of respondents not providing a scoring. “No awareness” means that the dis-services are not easily visible, and the respondents may not be aware of them. “No occurrence” means that the respondents were able to observe the dis-service but did not observe it.

fruit trees (Fruit trees), which were grown in areas with relatively low population (Population) and closer to the city center (Distance to city). These less diversified cropping systems (negative correlation with variable ‘Diversified cropping system’) had high cultural value (Tourism), despite signs of dis-services of water and air pollution (Fig. 7).

3.7. Differences in agroecosystem (dis)services supply among village types

Village types are mainly different in landscape composition, i.e., the area proportion of different crop types (Table S3). Villages with widespread vegetable cultivation (Village type 5) had generally better provisioning services (Revenue), regulating services (Ladybugs, etc.), and supporting services (Earthworms, etc.). Villages with widespread cotton cultivation (Village type 6), had relatively lower services deliveries especially cultural services (Fig. 8a), but higher dis-services like hail damage (Fig. 8b). The delivery of agroecosystem services was close to average in village types with widespread cereal cultivation (Village types 3 and 4) (Fig. 8a). Farmers in village type 4 reported few natural habitats (Ditches and Field islets) and ladybugs (Fig. 8a), but more dis-services to agriculture (Wind) and more dis-services from agriculture (Water pollution) (Fig. 8b). In the diversified village type 7, dis-services were lower than in other village types (Fig. 8b).

4. Discussion

Farmers in Quzhou County perceived that regulating and supporting services were in poor condition or declining due to groundwater depletion, scarcity of natural habitats and loss of natural enemies, but they did not think pests were serious (Fig. 3, Figs. 4 and 6). Dis-services generated by agricultural production were observed by only few farmers (Fig. 6). We investigated the relationship between perceived agroecosystem (dis)services and biophysical and socioeconomic conditions in villages, and found that the perceived agroecosystem (dis)services supply was associated with cultivated crop types and the distance to the city center (Fig. 7). Among different village types in Quzhou, the villages with a large proportion of fiber crops (cotton) had the worst performance in terms of agroecosystem (dis)services supply among the village types, while the villages with more vegetable production performed better (Fig. 8).

4.1. What can be learnt from farmers’ perception of agroecosystem (dis) services?

Farmers in Quzhou county perceived a scarcity or a decline of regulating and supporting services. Consistent with previous studies, the farmers participating in this study acknowledged several issues, including ground water depletion (Zhao et al., 2019), climate warming and heat stress (Chen and Whalen, 2016; Wu et al., 2021), a scarcity of

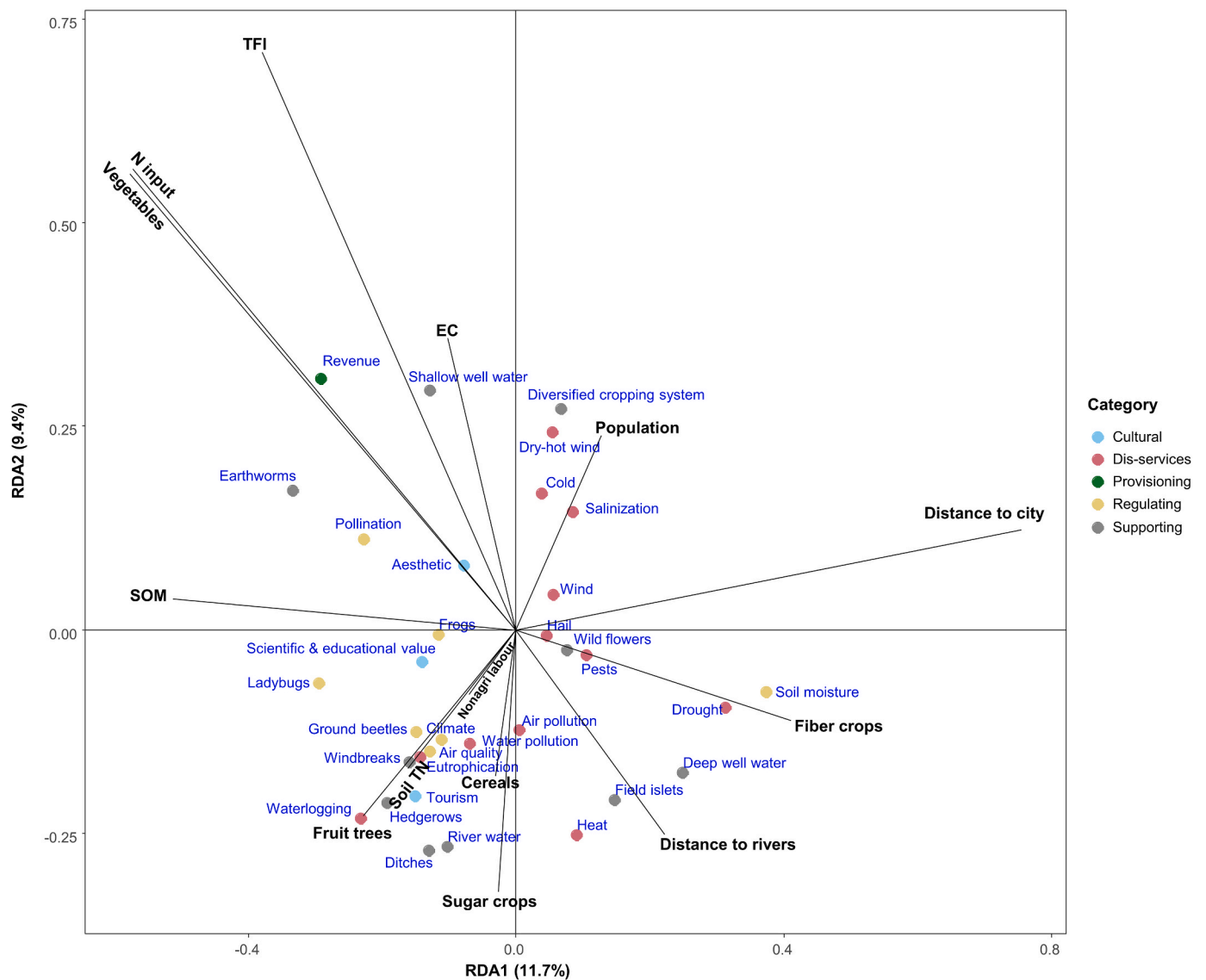


Fig. 7. Redundancy analysis of associations of village characteristics and the provisioning of agroecosystem services and dis-services. Dots with different colors represent different categories of agroecosystem services and dis-services. Lines with labels represent explanatory variables (Village characteristics). Definitions of each variable can be found in Table 1). Longer vectors indicate that a variable more strongly drives the variation of agroecosystem (dis)services among the villages. Angles between variables reflect their correlation (line to line, line to dot, and dot to dot) (Borcard et al., 2011). Lines or dots pointing in the same directions have a positive relationship.

natural habitats (Li, 2014) and loss of biodiversity in the simplified agricultural landscape (Liu et al., 2006, 2010).

Although the input of pesticides has been decreased by 35% and fertilizer was reduced by 9% from 2015 to 2020 in Quzhou County (NBSC, 2015–2020) under the “Zero Growth of Chemical Fertilizer and Pesticide Use Policy” that started in 2015 (Jin and Zhou, 2018), farmers still perceived a decline in the presence of natural enemies during this period. However, the actual dynamics between pests and natural enemies and the effects of (scarcity of) natural habitats would be hard to predict for farmers (W. Zhang et al., 2016). Stronger reductions in chemical inputs and the frequency and severity of mechanical operations (Bianchi, 2022) and the development of semi-natural habitats (Ricci et al., 2019) may be required to turn the agricultural landscape into “source habitats” for natural enemies (Dreyer and Gratton, 2014) to provide more effective pest control.

Tombs in traditional croplands can serve as less-disturbed habitat islets that could act as hotspots of biodiversity in the agricultural landscape of Quzhou (Li, 2014). They can function as refuges to host

farmland organisms and potentially natural enemies (Li, 2014; Xiao, 2014). The tombs hamper mechanical operations but have an important cultural value. Farmers indicated that removing the tombs from the fields for the purpose of land consolidation would cause resistance and resentment among farmers, especially among older people. This indicates the importance of considering the cultural values embedded in agricultural landscapes in the assessment and planning of landscape changes.

A striking result is the reported absence of dis-services from agricultural production. Most farmers reported to be unaware that overuse of agronomic inputs could cause ecosystem dis-services like air and water pollution. However, other research conducted in Quzhou county reveals that more than half of the farmers in the case study area extensively used pesticides, particularly in vegetables and orchards production (Mu et al., 2022a). This widespread pesticide usage led to health risks for farmers due to the presence of pesticides in indoor dust, which can be considered as a form of dis-service (Mu et al., 2022b). So why are dis-services reported to be so low? Three potential explanations

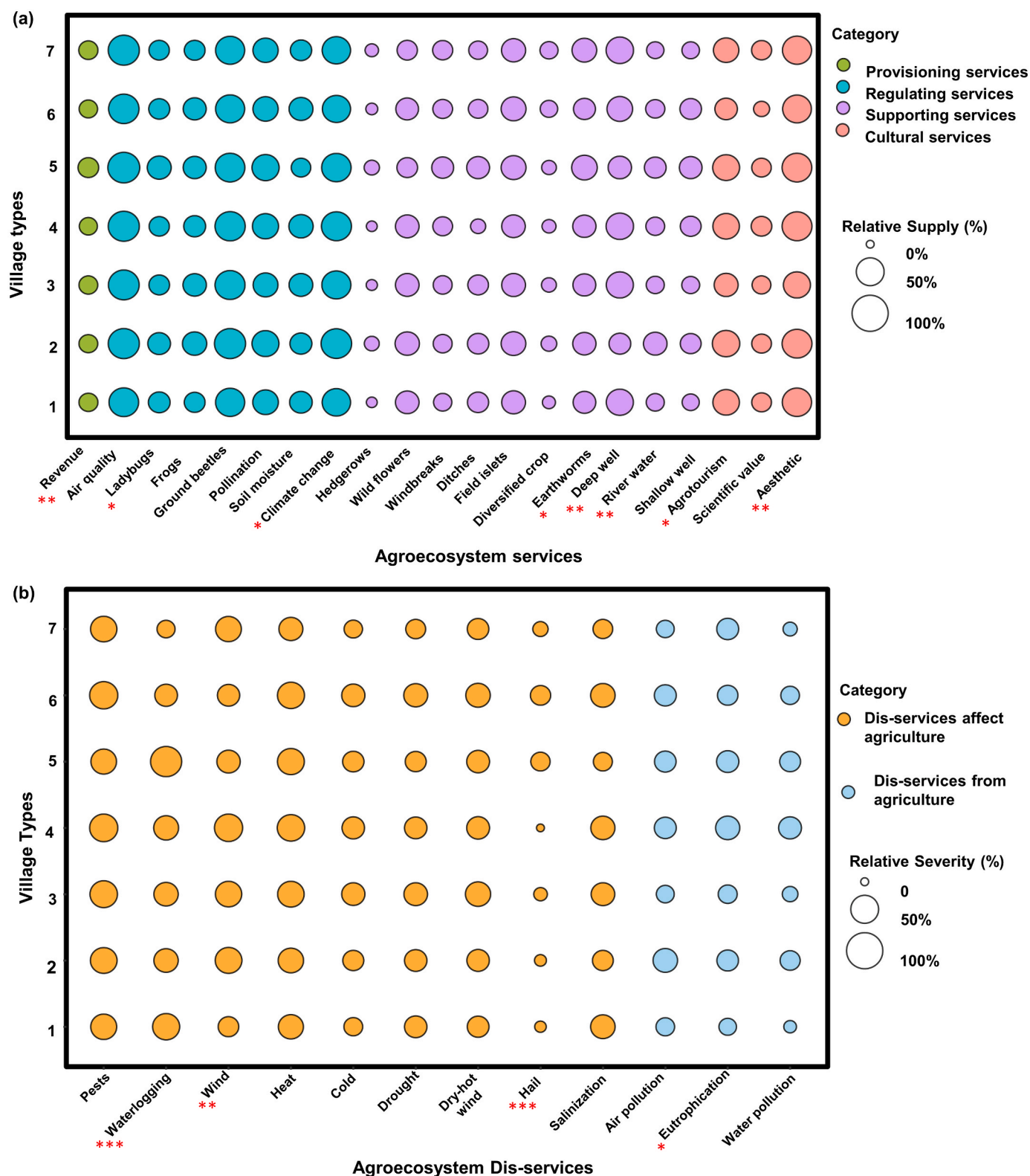


Fig. 8. Differences in ecosystem services (a) and dis-services (b) among village types. Village type 1 had relatively widespread fruit trees and was nearest to the city center. Village type 2 had more widespread cultivation of sugar crops. Land use in village type 3 was dominated by cereals and this village had high population density. Village type 4 was also cereal dominated. Village type 5 was vegetable dominated. Village type 6 had a large area proportion of cotton and was furthest to the city center. Village type 7 was relatively diversified. See Table S4 in supplementary material for additional information.

come to mind. First, many ecosystem (dis)services are difficult to observe even for knowledgeable and experienced observers (Lima and Bastos, 2019), so farmers are simply unable to perceive and interpret them (cf. Pan et al., 2021). Input suppliers were important sources of information of agricultural inputs application for farmers in the absence of a local extension system (Jin et al., 2015). However, those suppliers may not be able or willing to fully inform the farmers about the potential harm of chemical inputs, and some of them may amplify the recommended amount of pesticide use to gain profit or trust (Zhang et al., 2015). Second, many farmers in our survey attributed pollution to industrial and household waste, suggesting that farmers may not want to perceive or may not want to admit their contribution to environmental problems. A well-known caveat of self-reported assessment is that respondents tend to give socially desirable answers and/or show self-deception (Milfont, 2009). There could also be strategic motives involved to keep problems ‘under the rug’, for example anticipation of stricter regulation. Third, we may also observe a ‘shifting baseline syndrome’ (Soga and Gaston, 2018) where farmers are over time exposed to increasing levels of pesticides, and those levels are then perceived to be perfectly normal and seen as benign. Our research was not designed to disentangle or test the relevance of these three channels, but investigating the mechanisms and motives of underreporting of dis-services is certainly a fruitful avenue for further research.

4.2. Agroecosystem (dis)services and village characteristics

Previous studies have considered the effect of landscape features on the perceptions of ecosystem services, for example, non-crop habitats (Zhang et al., 2018), forest and agroforest (Muhamad et al., 2014), and multiple natural and semi-natural landscapes (Zoderer et al., 2019). In the context of agricultural landscapes, we are not aware of any research on the relations between people’s perceptions and landscape composition. This study provides evidence that the crop type composition in landscapes is strongly associated with farmers’ perception of agroecosystem (dis)services (Fig. 7). Socioeconomic and demographic factors also determine the ecosystem (dis)services and the perceptions, e.g., location advantages (Zhang et al., 2021), population density (Fei et al., 2016), and type of stakeholders (Bernues et al., 2016). Similar to previous studies (Wang et al., 2021), we also found that the distance to the city is associated with agroecosystem (dis)services delivery.

Villages with predominantly market-oriented crops (vegetables, fruits, Stevia) and located closer to the city center tended to have better soil and availability of irrigation. These advantages and the vicinity of markets provide those villages more options to develop cash crop production. Agrotourism also benefits villages with more fruit orchards and near cities by increasing their profit through activities like fruit-picking or flower-viewing. Previous studies indicated that excessive amounts of pesticide were used in vegetables and orchards production in Quzhou, which imperiled the pollinators (Mu et al., 2022a). Our survey was designed to measure farmers’ assessment of insect pollination services provided by naturally occurring insects and managed honeybees. The results showed the farmers in these vegetable-dominated villages perceived effective pollination. From our field visits, we know that farmers in these villages may nevertheless use hand pollination (e.g., in melons) indicating that they still invest in measures to assure their yield. Such pollination measures require additional input and investment compared to natural insect-induced pollination, and assessing the socioeconomic impacts of those seems like a natural avenue for further research.

Remote villages tend to utilize a larger proportion of the area for cotton cultivation, which may result from the fallow policy that was introduced in 2015. The government encouraged farmers to grow single season cotton instead of winter wheat-summer maize double cropping to reduce groundwater depletion. Farmers adopting fallow policies received subsidy equivalent to the profit of one season grain production. Despite the extensification of land-use due to the following, which is

expected to deliver more services (e.g., save more underground water), these farmers perceived more dis-services compared to beneficial services.

4.3. Limitations and outlook

Perception-based evaluation can be an effective scanning tool to elucidate locally relevant constraints, challenges and patterns of observations and beliefs regarding agroecosystem (dis)services (Chen et al., 2018; Smith and Sullivan, 2014). The survey-based assessment of agroecosystem (dis)services could be used to rapidly identify issues and challenges experienced by stakeholders in relation to agricultural production in a bottom-up approach (Dawoe et al., 2012), but there are important caveats. Perceptions are inherently subjective and dependent on the respondents’ ability and willingness to acknowledge problems (Bennett, 2016). In addition, multiple factors like visibility of (dis)services, knowledge, experience, biophysical environment, personality traits, demographic characteristics, etc. also make the interpretation of perception-based agroecosystem (dis)services assessment complex (Chen et al., 2018; Lima and Bastos, 2019; Martín-López et al., 2012; Zhang et al., 2018; Zoderer et al., 2019). Knowing to what extent stakeholders can precisely characterize the real agroecosystem (dis)services and their trend over time is challenging. The results from perception-based evaluation of agroecosystem (dis)services can provide valuable bottom to up information, but policymakers should always be mindful of their limitations.

Quzhou is a representative county in the North China Plain, but perception-based evaluation is still context-dependent. As Quzhou became a test area for China’s policy for Agriculture Green Development in 2020, the perception of ecosystem services in this area could change with time. Therefore, quantitative assessment with measurable (proxy) indicators is needed. Survey-based assessments are most valuable when complementing – rather than substituting – an indicator-based biophysical assessment. The interpretation of our study results is to some extent constrained by the absence of ground-truthing data. Nevertheless, the current survey results set a useful baseline for future monitoring and measurement programs, and for evaluating the temporal development of stakeholder perceptions.

There are several important considerations to bear in mind when comparing perceptions with biophysical assessment through ground-truthing in future research. Firstly, relying solely on biophysical evaluations may not adequately capture the subjective cognitions and experiences of individuals concerning cultural services. Secondly, respondents may not fully align their understanding of certain (dis)services with the definitions used in biophysical assessment. Thirdly, some (dis)services, such as air pollution, are invisible and require a theoretical understanding to fully comprehend their impact. In such cases of people’s non-recognition of certain (dis)services, ground-truthing can yield more reliable results. In conclusion, the process of ground-truthing holds value for verifying perceptions of agroecosystem (dis)services. However, it is crucial to approach ground-truthing with careful consideration of the unique characteristics of each (dis)service and to be mindful of potential reasons for discrepancies between people’s perceptions and biophysical assessments. By addressing these factors during ground-truthing, researchers can obtain more reliable and comprehensive insights into the true nature of agroecosystem (dis)services.

The results in this study are based on a sample of interviewees consisting of 187 smallholder farmers, 35 village heads and 45 leader farmers. We interviewed all the village heads in the sample of 35 villages, all the leader farmers that could be reached, and a proportion of the smallholders accounting for 100–800 households per village. Results presented are for the pooled data, but some differences between the three groups of interviewees were identified (Table S5). We chose to analyze the pooled data of the whole sample because on the whole, the three farmer groups of interviewees gave scores that are consistent in the

sense that they would qualify as low, medium or high across all the groups, even in the cases in which significant differences between the groups were detected (Table S5). While these differences are acknowledged, the analysis of this paper is not focused on comparing different stakeholders but on an overall assessment of ecosystem services provisioning in the study area. For an assessment, the point of view of leader farmers and village heads cannot be ignored because they are generally well informed on the local situation and in any future societal discourse, such stakeholder will tend to have a voice. Overall, the ratings of smallholders (187 interviewees out of a total of 267 interviewees; 70%) still determine to a large extent the overall outcome of the survey. Supplementary material provides Figs. S5–S8, based on the smallholders only, showing broadly the same patterns as those represented in the main text, so the conclusions drawn are robust to this selection of the sample.

The redundancy analysis (RDA) explained only a small proportion of the variance in this study. RDA can only reveal linear relationships between explanatory factors and agroecosystem (dis)services, while for the ecological processes underlying agroecosystem (dis)services provisioning non-linear relations could be expected. Therefore, the RDA cannot address the complex nature of the socio-ecological systems and processes occurring at multiple temporal-spatial scales. Nevertheless, RDA can still give an impression of the direction of relations since it captures the key factors that influence the agroecosystem (dis)services and allows to visualize the relationship among the variables. The robustness of the relationships needs to be further tested at multiple temporal and spatial scales.

The village-based analysis suggests that agroecosystem (dis)services are context-specific and should therefore be linked to socioeconomic and biophysical factors that are likely to vary between villages in the same county or city. Accordingly, Chinese government policy proposed in 2019 encourages using a village typology for rural planning and development (Ministry of Agriculture and Rural Affairs, 2019). Moreover, participation of farmers and other local actors in the planning and development process is stimulated. In that sense, the farmers' perception of agroecosystem (dis)services could be embedded within these wider policy development processes. This would create an interface between the top-down and bottom-up planning approaches which can help to make policy implementation more effective (Martínez-Sastre et al., 2020; Pradhan et al., 2017). Farmers could benefit from their involvement as they would receive more professional knowledge, information, and guidance for nature-positive farming practices. This could be combined with existing interface platforms like the science and technology backyards (STBs), which support agricultural innovation within villages (J. Li et al., 2022).

5. Conclusions

Here we describe the current state and recent trends in agroecosystem services and dis-services perceived by farmers in the simplified agricultural landscapes of Quzhou county in the North China Plain. The study shows that while Quzhou exhibits relatively good provisioning services, farmers perceived a scarcity and vulnerability of regulating and supporting services and had limited awareness of dis-services. The lack of awareness of dis-services is surprising, as several dis-services have been documented on the basis of quantitative measurements in other studies. This suggests that farmers' perception can complement but not substitute agroecosystem services assessments based on biophysical quantification. The farmers' perception can be used to effectively search for the knowledge gaps of ecosystem services assessment and where extension and awareness raising are needed. Verification of farmers' perceptions by ground-truthing could further contribute to having a baseline for monitoring the progress of 'Agriculture Green Development' in the future. The findings of the relations between perceived supply and village characteristics emphasize the need for targeted interventions and policies to enhance agroecosystem service

provision at the village scale. Stimulating nature-positive farm practices and improving the agroecosystem services requires consideration of specific socioecological conditions of different villages and engagement of different stakeholder groups.

Credit author statement

Jiali Cheng: Investigation, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. Zhan Xu: Data curation, Formal analysis. Zhengyuan Liang: Investigation, Data curation. Fan Li: Investigation, Resources, Writing – review & editing. Wen-Feng Cong: Supervision, Writing – review & editing, Funding acquisition. Chaochun Zhang: Supervision, Resources. Lei Song: Investigation, Resources. Chong Wang: Data curation, Resources. Fusuo Zhang: Supervision, Resources, Project administration. Andries Richter: Supervision, Writing – review & editing. Wopke van der Werf: Conceptualization, Supervision, Writing – review & editing. Jeroen C.J. Groot: Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2023.119060>.

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